Beyond Single-Purpose Apps: Investigating the relevance of mobile usability heuristics in modern multifunctional app design through a fitness case study

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1. Project background and purpose

1.1. Introduction

Mobile applications have developed over the past several years from straightforward, one-use tools to sophisticated, multipurpose platforms that help users accomplish numerous related tasks from a single interface. Greater efficiency and convenience are offered by this evolution; However, usability and user experience also face unforeseen challenges. Traditional mobile usability heuristics, such as those put forth by Nielsen and Gerhardt-Powals, were created in a time when apps usually had a single, obvious purpose. Because of this, their direct application to contemporary multifunctional designs has not been fully reassessed in light of user expectations, visual complexity, and modern interaction patterns.

To complete related tasks, many mobile users rely on several single-purpose apps, which leads to disjointed workflows and uneven user experiences. There are potential and design problems associated with the growing popularity of multifunctional apps, or apps that combine multiple tools into a single platform.

The ultimate aim of this experiment is to find out if these established heuristics still work well for multipurpose mobile applications. In order to help the design of integrated, feature-rich applications, it will concentrate on *determining where existing heuristics work well, where they could prove ineffective, and whether new or modified guidelines are required.*

The project will combine theoretical investigation with practical implementation to analyse this. As a controlled case study, a multipurpose fitness companion app with functions including notetaking, workout tracking, and rest timers will be created. Heuristic evaluation will be based on this prototype, allowing for the examination of usability problems unique to multifunctional settings.

While it is true that paradigms for mobile engagement are continuously shifting, emerging technologies like wearable interfaces, augmented reality (AR), and voice-based systems are having a growing influence on user experiences, this does not diminish the relevance of investigating mobile usability today. The fundamentals of usability are not limited to any one technology, but rather to the more broad concepts of human-computer interaction. Both future interaction models and legacy applications can benefit from the adaption of these concepts based on the knowledge gathered from assessing heuristics in contemporary mobile contexts. In this way, it offers a basis for these ideas to develop with technological advancement rather than being rendered obsolete by it.

Despite all of the careful consideration that went into this study, exploratory research always involves some degree of uncertainty. Like any investigative study, there's a chance that outcomes will differ significantly from the initial expectations. For instance, the prototype might not be completed to the desired level, or the evaluation might show conclusions that are unclear or unfavourable. Such results would not, however, indicate failure; rather, they would offer insightful information on the drawbacks of incorporating conventional heuristics into contemporary app design and guide suggestions for further research. The project's focus on critical analysis and reflection guarantees that significant conclusions regarding the continued applicability of usability heuristics in changing mobile contexts may be made regardless of the outcome. Later in this report, strategies for ensuring this project is completed to the level promised are discussed.

By assessing whether current usability heuristics are still applicable in the current context or whether the emergence of multifunctional apps necessitates a change in the definition, testing, and measurement of usability, the project ultimately aims to contribute to ongoing discussions in software engineering and user experience design.

1.2. Objectives

This project's main goal is to assess the validity and continued relevance of conventional mobile usability heuristics in the context of designing multipurpose applications, using empirical evaluation.

To achieve this, the project will pursue the following *primary* objectives:

- Establish the theoretical foundation for the research by ***conducting a literature review*** on mobile usability heuristics and their historical application by *week 7.*
    - Identify key principles and any existing research that questions their modern relevance.
    - Focus on established frameworks (e.g., Nielsen's, Gerhardt-Powals').

- ***Examine a sample of multipurpose mobile applications*** to determine how they adhere to or depart from conventional heuristic concepts by *week 11*.
    - Analyse 2-4 mobile applications, ideally these will be thematically and functionally different (e.g. utility, productivity, social).
    - Develop a comparative table for mapping heuristic compliance.

- ***Design a multipurpose fitness application prototype*** by integrating gym-related capabilities into wireframes and user interface mock-ups *by week 12*.
    - Create design artefacts with an emphasis on user flow, layout and navigation.
    - Include at least three functions (e.g., timers, workout logging, and note-taking)

- ***Build the designed prototype*** using the chosen frameworks and technologies, incorporating the planned features, to produce a usable, multipurpose fitness app *by week 17*.
    - Developed using the wireframes previously drawn up.
    - Ensure product offers a suitable setting for user usability testing.

- ***Conduct user-based usability testing*** in order to gather quantitative data to assess accuracy of heuristics gathered in literature research *by week 20*.
    - Pick appropriate testing techniques to get empirical evidence such as A/B testing or timed task performance.
    - Pick tester sample size of at least 3-5 people.

- ***Analyse and synthesise the test data*** to ascertain whether the current usability guidelines are still applicable to the design of multifunctional mobile applications *by week 23*.
    - Compare user data with heuristic predictions.
    - Produce a written evaluation identifying consistencies and discrepancies.
    - Synthesise findings to answer whether or not "Contemporary heuristics are relevant and accurate for multifunctional mobile application development?".

The following *secondary* objectives are not necessary to answer the question proposed by the main goal of this study; However, their inclusion would increase the quality of the concluding evaluation significantly and extend the value of the project.

- Propose refinements or extensions to current usability heuristics.
- Revise the prototype to reflect these improvements and showcase enhanced usability.
- To compare the usability metrics before and after refinement, undertake a secondary user test.
- Publish the prototype onto a mobile application digital distribution service (e.g. Android play store, Apple app store)

1.3. Scope

This project will focus primarily on the usability and design evaluation of multifunctional mobile applications, paying particular attention to the use of well-known usability heuristics within these contexts. Through user-based usability testing, the practical component—a multifunctional fitness app prototype—will act as a controlled environment for evaluating and analysing heuristic relevance.

The scope **includes:**

- Research and critical review of existing mobile usability heuristics with practical examples of open to use applications.
- Design and development of a prototype multifunctional fitness app integrating multiple tools.
- Analysis of chosen, common heuristics and user-based testing using a small, achievable sample size.
- Evaluation of findings in relation to relevance of established heuristic principles.

The scope **Excludes:**

- Large-scale deployment or fully realised commercial release.
- Backend data synchronisation, cloud services or server-side development.
- Full-scale longitudinal user study.

1.4. Deliverables

In order to answer the main research question about the continued relevance of mobile usability heuristics in contemporary multifunctional app design, this project will generate a variety of research, practical and evaluative deliverables.

The following are the anticipated deliverables:

- Research Report: A thorough written report that examines current usability guidelines and how well they work with multipurpose mobile applications. A review of the literature, an analysis of the case study, and a critical discussion of the results will all be part of this.

- A functioning mobile application that combines several gym-related elements (such as a workout tracker, rest timer, and note-taking capabilities). For empirical usability testing, the prototype will act as a controlled setting.

- Evaluation Dataset: A/B comparisons, task completion times, and user preference feedback are among the quantitative and qualitative data gathered by user-based usability testing. The effectiveness of established heuristics in explaining observed usability patterns will be evaluated using this dataset.

- Heuristic Recommendations: A more accurate or expanded set of usability heuristic guidelines designed for the design of multipurpose mobile apps, derived from the findings of the evaluation and insights from the literature.

- Project documentation: Supporting artefacts that show the development process, evaluation techniques, and logic behind design choices, such as test protocols, design mock-ups, and analytical materials. Though this will likely be a part of the study report.

These deliverables will show whether current usability criteria are still applicable or need to be modified for modern multifunctional mobile contexts. They will also offer theoretical insight and practical validation

1.5. Constraints

The project will work under a number of technical and practical limitations that could affect how it develops and is evaluated.

- Platform Limitation: Because of time constraints, development experience, and avail able resources, development will likely be limited to the Android operating system. Implementing cross-platform software (like iOS) is outside the present project scope due to a lack of apple hardware.

- Time and Resource Restrictions: The project needs to be finished by the honours stage module's academic deadline.
  This restricts the possibility of conducting extensive testing, creating intricate features, or recruiting a big number of users.

- Evaluation Restrictions: A limited user group will participate in usability testing, which will emphasise both qualitative and quantitative task-based assessment.
  This may limit the conclusions' statistical generalisability, but it will still offer valuabl e information about the applicability of the heuristic.

- Logistical and Ethical Restrictions: All user testing will need voluntary involvement and ethical permission. As a result, this may again contribute as to why there might not be as many people available or diverse.

- Financial limitations: This study is being done with a very small budget, paid services or subscriptions will likely be out of the question, this may impact quality of product/data.

- Power of conclusion limitations: All these restrictions may mean that ultimately the resources available cannot answer the question raised beyond a shadow of a doubt. This may impact the reliability of data gathered or the credibility of the entire document.

To guarantee that the task remains feasible while still fulfilling the objectives of the study and yielding significant, reliable results, these limitations have been acknowledged and taken into account in the project plan.

1.6. Assumptions

The planning and effective completion of this project are predicated on several assumptions:

- Validity of Current Heuristics: It is believed that well-known usability heuristic frameworks, like those put out by Nielsen and Gerhardt-Powals, still offer a reliable and pertinent starting point for evaluating contemporary mobile applications. In multifunctional app situations, this assumption enables a meaningful comparison between observed usability outcomes and conventional heuristic concepts.

- User Familiarity: Participants chosen for usability testing are presumed to have a rudimentary understanding of touchscreen interaction and mobile applications. This guarantees that usability issues found during testing are related to design elements rather than a deficiency in mobile device user experience.

- Tool Reliability: Throughout the project, it is anticipated that development and testing and u sability testing software will operate dependably and without experiencing major technical d ifficulties.

- Availability of volunteers: It is anticipated that there will be enough volunteers available to fi nish usability testing sessions in the allotted period, and that ethical approval for such testin g will be obtained quickly.

These presumptions give the project a solid basis and will be examined on a regular basis to make sure that any modifications to the situation are recognised and handled properly.

2. Project rationale and operation

2.1. Project benefits

The project contributes to understanding how usability heuristics evolve alongside mobile technology trends. The findings could benefit UX designers, software engineers, and researchers by clarifying which established principles still hold value and where modern app design requires new considerations.

Currently the efficiency and reliability of mobile heuristics is uncertain, developed at different time, they may prove not applicable or appropriate in the modern context. This project, if successful, should aid in this discussion, deducing whether selected heuristic guidelines still apply. If this question is answered, fresh or altered guidelines may be written, fixing/ modernising those selected.

2.2. Project operation

The project will adopt a ***hybrid methodological approach*** that combines structured planning and iterative engineering. At the macro level, the report/project will follow ***waterfall principles*** due to the necessary chronological ordering of objectives. For example, analysis of test data cannot be completed in tandem or before prototype development. This structured sequencing ensures logical progression.

Within this overarching framework, the ***agile methodology guided by scum principles*** will be employed into the prototype's design and development phases. The development process will be partitioned into smaller, more manageable sprints; These sprints will each individually focus on a singular feature/aspect such as workout tracking. Each sprint will adhere to an ***iterative*** cycle of planning, execution, assessment and improvement based on feedback from both the project supervisor and internal testing. These methodologies should encourage adaptability, flexibility and continuous improvement.

Day to day project management will involve accountability and maintaining momentum as guided by the Gannt chart and schedule. Realistic scope and practical time frames will result in a project that does not derail due to lack of foresight or overconfidence in personal resources. Weekly supervisor meetings will also aid in the monitoring of project progress and accountability while also providing vital on-going feedback. Version control software like git will be used to record project progress and preserve code history. As well as this, tools such as Trello will house sprint backlogs.

2.3. Options

Currently, a number of potential categories have been identified that will need to be further examined as the project develops. After preliminary research and prototype preparation, the ultimate choice of tools, frameworks, and processes will be made to make sure they complement the project's goals, limitations, and resources.

Development Environment Options: A variety of platforms and frameworks, such as Android Studio, Flutter, or React Native, will be taken into consideration when creating the prototype. The ultimate decision will be based on a number of criteria, including usability testing acceptability, compatibility with planned features, support resources, and ease of use.

Testing and assessment Techniques: There are several testing techniques that can be used, including think-aloud research, task-based usability testing, A/B testing, and heuristic assessment. The particular approach—or combination of approaches—that best gathers quantitative and qualitative information pertinent to the research topic of the study will be selected.

Project Management Approach: A variety of project management techniques, such as Agile, Scrum, and Waterfall, will be examined. The best approach or hybrid model will be chosen in accordance with supervisor input, workflow requirements, and time limitations.

In order to ensure that the tools and methods selected support the study's research and practical development components, final decisions within each of these categories will be made early in the project, led by preliminary research findings and iterative conversations with the project supervisor.

2.4. Risk analysis and mitigation

Risk	**Likelihood**	**Impact**	**Mitigation Strategy**
Time Management and Scheduling *– Delays in completing research, development, or testing tasks due to underestimating workload or unforeseen complications.*	Medium	High	Maintain a detailed Gantt chart and weekly task reviews. Prioritise critical milestones first (e.g., literature review, prototype development) and adjust non-critical tasks if delays occur.
Technical Difficulties *– Unexpected issues with chosen development tools or frameworks causing delays.*	Medium	Medium	Begin early experimentation with shortlisted tools. Maintain backup options (e.g., switching from Flutter to Android Studio) and document setup procedures.
Limited Participant Availability *– Difficulty recruiting enough users for usability testing due to time or ethical constraints.*	Medium	Medium	Start ethical approval and recruitment planning early. Use a small but representative user sample and consider remote or simulated testing if necessary.
Data Reliability *– Inconsistent or incomplete usability test data (e.g., user fatigue or misunderstanding tasks).*	Medium	Medium	Pilot test evaluation procedures to refine task clarity and ensure test consistency. Use both quantitative and qualitative measures to validate findings.
Scope Creep *– Attempting to add unnecessary features to the prototype, leading to delays or diluted focus.*	Medium	High	Adhere to the defined project scope. Implement only essential functions for usability evaluation (e.g., timers, notes, workout logging).
Ethical or Legal Issues *– Delays due to ethical approval requirements or mishandling of user data.*	Low	High	Submit ethics form early. Follow GDPR guidelines for anonymised data storage and disposal.
Hardware or Software Failure *– Loss of data due to device failure or corrupted project files.*	Low	High	Use version control (GitHub) and cloud backups for all project files. Maintain regular checkpoints and redundant storage.
Negative or Inconclusive Results *– User testing may reveal that heuristics are not strongly applicable to multifunctional contexts.*	Medium	Low	Acknowledge that inconclusive results still provide valuable insights. Ensure findings are critically analysed and reflected upon rather than dismissed.

2.5. Resources required

The project will need standard productivity and data analysis software, as well as access to an appropriate development environment for creating mobile applications. It will require a mobile device or emulator for testing, as well as a personal computer that can run development and design tools. The project supervisor and a small group of usability testers will be part of human resources. No non-standard hardware or paid services will be required.

2.6. Ethical and legal considerations

While this project involves very limited ethical risks, thought will be put into consideration for research involving human participants as to adhere to the university's ethics guideline provided. Usability testing will require fully informed consent from each subject with the right to remove themselves from the study at any time. This withdrawal from the project will also necessitate the destruction of all data relating to that participant with no consequences to the person/persons. No sensitive personal data will be collected, and all responses will remain anonymous. The participants will be fully briefed on the study's purpose, testing procedures and data taken.

All participant data will be securely stored in a database accessed only be the head of project. This data will be only used strictly for research purposes in compliance with GDPR requirements.

2.7. Commercial considerations

Though the focus of this initiative is purely academic, the results may have commercial value. The idea of combining several single-purpose fitness tools with different functions into a single multipurpose mobile application is in line with the market's growing trend towards integrated digital ecosystems. By finding principles that increase user satisfaction and engagement, the evaluation of usability heuristics could provide valuable insights for future commercial app design techniques.

The project itself will incur minimal costs in terms of development resources, software, and hardware. There are no plans for direct monetisation or income generation. The prototype may be the foundation for additional commercial development or cooperation outside the purview of this study if great usability and user appeal is shown. In addition to this, this project may act as proof of the publisher's competency in the subject area of mobile application heuristics, this may open doors of consultancy.

3. Project methodology and outcomes

3.1. Initial project plan

3.1.1. Tasks and milestones

1. Complete final draft of the project definition document.
2. Produce a clear refined specification of what needs to be done.
3. Consider different technologies for achieving the aims of the project and select the most appropriate.

--Planning phase over--

4. Explore the history of the types of user interface.
5. Investigate the history of user interface workflows.
6. Conduct a literature study and review on mobile application usability heuristics (select heuristics to analyse) and multifunctional application design.
7. Identify case studies of both single and multi-function mobile applications and analyse against chosen focus heuristics.

--Literature review phase over--

8. Define requirements for the fitness app prototype.
9. Create wireframes and user interface mock-ups ranging from low to high fidelity.
10. Select frameworks and languages for development with justified reasoning and gain competency in technologies.
11. Develop prototype app including only the core features.

--Prototype development phase over--

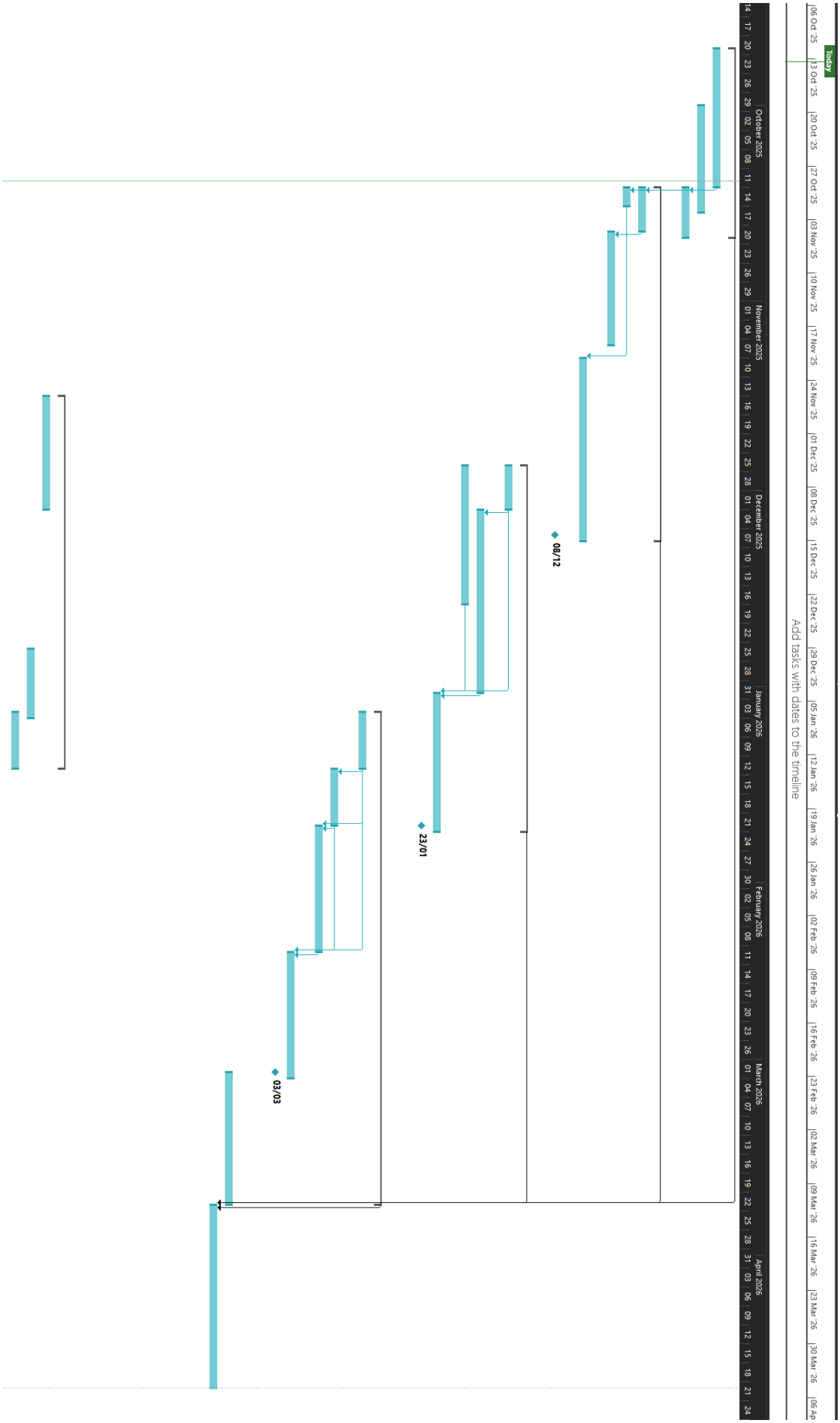
12. Conduct a secondary literature review focusing on heuristic evaluation protocols (use this time to research any principles not covered accurately enough during first literature study).
13. Select testing techniques, create test plan, recruit participants and set up secure storage for test data.
14. Conduct heuristic evaluation and collect feedback.
15. Analyse results and propose heuristic relevance/refinements.

--Heuristic evaluation phase over--

16. Finalise final report and project evaluation.

3.1.2. Schedule Gantt chart

	Task Mode	Task Name	Duration	Start	Finish	Predecesso
1		**Prep**	**22 days**	**Mon 22/09/25**	**Tue 21/10/25**	
2		Complete PDD	16 days	Mon 22/09/25	Mon 13/10/25	
3		Refine Specification	13 days	Wed 01/10/25	Fri 17/10/25	
4		Consider different design and development technologies	6 days	Tue 14/10/25	Tue 21/10/25	2
5		**Literature review**	**40 days**	**Tue 14/10/25**	**Mon 08/12/25**	
6		History of UI	5 days	Tue 14/10/25	Mon 20/10/25	2
7		History of workflows	3 days	Tue 14/10/25	Thu 16/10/25	2
8		Literature review of mobile usability heuristics and design	14 days	Tue 21/10/25	Fri 07/11/25	6
9		Identify sample of case studies and analyse	21 days	Mon 10/11/25	Mon 08/12/25	7
10		Literature review complete	0 days	Mon 08/12/25	Mon 08/12/25	
11		<New Task>				
12		**Prototype development**	**42 days**	**Thu 27/11/25**	**Fri 23/01/26**	
13		Define requirements of prototype	5 days	Thu 27/11/25	Wed 03/12/25	
14		Create wireframes	21 days	Thu 04/12/25	Thu 01/01/26	13
15		Select frameworks and gain competence	16 days	Thu 27/11/25	Thu 18/12/25	
16		Develop prototype	16 days	Fri 02/01/26	Fri 23/01/26	13,14,15
17		Prototype successfully developed	0 days	Fri 23/01/26	Fri 23/01/26	
18						
19		**Heuristic evaluation**	**56 days?**	**Mon 05/01/26**	**Mon 23/03/26**	
20		Literature review of heurstic evulation	7 days	Mon 05/01/26	Tue 13/01/26	
21		Create test plan	7 days	Wed 14/01/26	Thu 22/01/26	20
22		Conduct heuristic evaluation and collect feedback	14 days	Fri 23/01/26	Wed 11/02/26	20,21
23		Analyse results	14 days	Thu 12/02/26	Tue 03/03/26	20,21,22
24		Heuristic evaluation complete	0 days	Tue 03/03/26	Tue 03/03/26	
25						
26						
27		-Buffer zone-	15 days?	Tue 03/03/26	Mon 23/03/26	
28		Project evaluation and report finalisation	21 days	Tue 24/03/26	Tue 21/04/26	1,5,12,19
29						
30						
31						
32						
33						
34						
35						
36						
37		**Foreign commitments**	**42 days**	**Sun 16/11/25**	**Tue 13/01/26**	
38		Other module ACW due	14 days	Sun 16/11/25	Wed 03/12/25	
39		SDI exam	7 days	Fri 26/12/25	Mon 05/01/26	
40		Buffer zone - Time to reflect -	7 days	Mon 05/01/26	Tue 13/01/26	


Today
06 Oct '25
13 Oct '25
20 Oct '25
27 Oct '25
03 Nov '25
10 Nov '25
17 Nov '25
24 Nov '25
01 Dec '25
08 Dec '25
15 Dec '25
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06 Ap
Add tasks with dates to the timeline
October 2025
November 2025
December 2025
January 2026
February 2026
March 2026
April 2026
08/12
23/01
03/03

** Note Gantt chart file is available in the GitHub repository, which may be easier to read **

3.2. Project control

A thorough task list and Gantt chart will be used to monitor progress, and they will be updated often to reflect finished milestones and any scheduling changes. In the event of delays or problems, supervisor meetings will offer chances for feedback, risk revaluation, and remedial action. To ensure that modifications are recoverable and traceable, development progress will be documented using version control software.

3.3. Project evaluation

Both the development process and the end results will be used to assess the success of the project. Schedule adherence, efficient risk management, and response to supervisor input will all be taken into account throughout the process review.

Analysing usability test findings will be the main focus of outcome evaluation in order to ascertain how well-established heuristics translate into the design of multifunctional mobile apps. To determine relevance, heuristic expectations will be matched with quantitative data, such as task completion times and user preferences.

If the project creates a working prototype, collects useful usability data, and provides well-supported findings about the applicability of current mobile usability heuristics, it will be considered successful.

4. References & Appendix

Heuristic evaluation tutorial:

https://www.ingenieriasimple.com/usabilidad/HeuristicEvaluation.pdf

Molich, R., and Nielsen, J. (1990). Improving a human-computer dialogue, Communications of the ACM 33, 3 (March), 338-348.

Nielsen, J., and Molich, R. (1990). Heuristic evaluation of user interfaces, Proc. ACM CHI'90 Conf. (Seattle, WA, 1-5 April), 249-256.

Nielsen, J. (1994a). Enhancing the explanatory power of usability heuristics. Proc. ACM CHI'94 Conf. (Boston, MA, April 24-28), 152-158.

Nielsen, J. (1994b). Heuristic evaluation. In Nielsen, J., and Mack, R.L. (Eds.), Usability Inspection Methods, John Wiley & Sons, New York, NY.

Ethics checklist available under 'Ethics checklist' in honour stage repository

Full Gantt chart is found under 'Gantt chart' in honour stage repository